

1. *Target User Population*

My target user population is made up of people who enjoy different forms of media content—both visual media (television and movies) and books—and who want a better way of getting recommendations for content. They especially want to be able to understand and control the recommendation process. The user population covers a broad swathe of people, ranging in age from teens to older adults.

The population is very broad, and does include me. (However, a number of the users I spoke to had different priorities than me.)

2. *Project Concept*

A description of your design solution and how it addresses an important problem faced by your target user population. It is fine — and often preferred — for your design solution to tackle a necessary subproblem rather than the problem at large. If this is the case for your project, you should explain the problem at large, what the subproblem is, and why it was important for you to focus on just that subproblem.

My project tackles a subproblem of a larger issue.

The larger problem faced by users is effective media-finding: users want to get recommendations for media titles from algorithms, determine whether they want to watch it, and then access the media. Among other inefficiencies, the recommendation process is opaque. Users cannot understand, visualize, or control why a website's recommendation algorithm is recommending a particular media title, even if that would help them find the media titles they want.

Other problems in the media-finding process are generally handled by other websites, like Netflix, IMDB, and Goodreads. However, there is no available source which allows users to clarify and control the recommendation process. Focusing on this subproblem lets my project serve an original and currently-unmet user need.

My design solution is fundamentally to create “links,” or connections, between different media titles. My site will use these connections as a source of media recommendations. That means that media titles will be recommended on the basis of criteria like ‘same author,’ ‘same director,’ ‘similar story,’ and so on. By recommending media on the basis of specific, named connections, I allow users to understand why they are being recommended a particular title. Users will also be able to choose which links they want to prioritize. This will allow them to customize why they are receiving particular recommendations.

3. *Challenge*

My project is a significant creative challenge. It also has a broad scope (especially earlier in my design process, before I had given additional focus to my problem statement) which made finding my ultimate design solution a challenge.

Firstly, my project has a very broad scope. After my initial contextual inquiry, I had thought to fix the many inefficiencies in the media-finding process. After my user interviews with the paper prototype, I honed in on the need that my users found most important—understanding and controlling their media recommendations—but this remains an extremely broad problem. In order to give users the ability to control their own recommendation process, my design solution has to acknowledge many different priorities from users of all different kinds. It also has to accommodate both books and visual media.

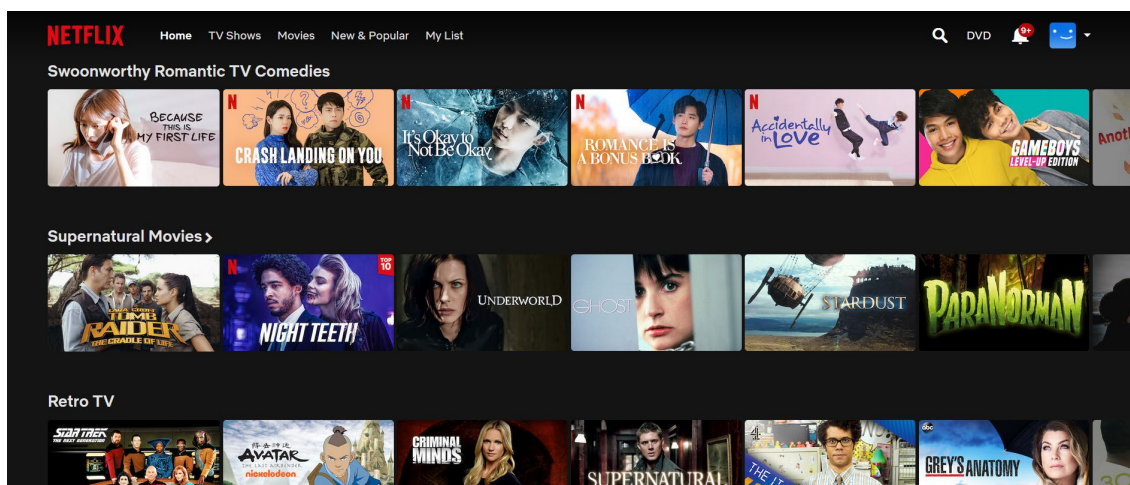
Secondly, the project is a creative design challenge. I had to reimagine the traditional model of media recommendations in a way which is innovative enough to engage users, intuitive enough to be usable, and which gives users understanding and control of the recommendation process. Figuring out how to let users control their recommendations was a significant challenge.

4. *Comparative analysis*

The majority of the users I interviewed primarily used streaming services to find recommendations. Many popular streaming services (Netflix, Hulu, Disney+, Peacock) use what I call the ‘Blockbusters’ model for recommendations. They show a horizontal array of media titles, similar to a shelf at the defunct video rental service Blockbuster, and each row is recommended based on a criterion chosen by the site, not the user.

These services do well by combining their central service—providing users with visual media that they can watch online—with recommendations. This creates a hub where users can find a show and immediately start watching it. They also do well by making the design similar to real-world designs—like shelves at a video store.

However, where they fail is in their lack of explanation and control. Users cannot choose how media titles are recommended to them, and they do not understand why these titles are recommended. Also, there is no way to get recommendations between streaming services—each service recommends only the titles that it owns, even if there are titles which the user would like better.



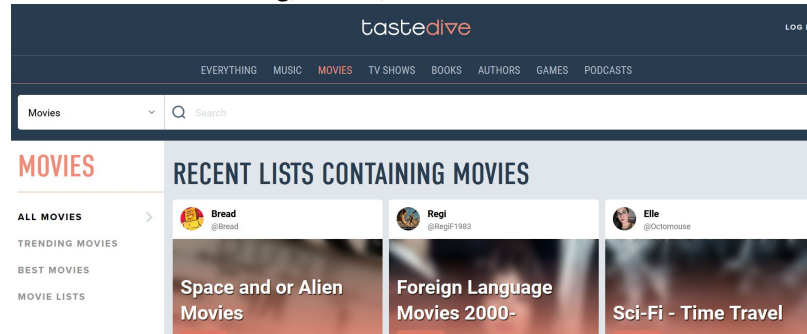
A page from the popular streaming service Netflix showing horizontal rows of visual media. The rows are labeled ‘swoonworthy romantic TV comedies,’ ‘supernatural movies,’ and ‘retro TV.’ There is no way to choose what options are shown.

On researching, there are also sites dedicated to finding recommendations, like TasteDive. TasteDive shows lists curated by individuals collecting different movies, TV shows, books, authors, games, podcasts, or music. It also shows which media titles of each type are ‘trending,’ and which are the ‘best.’ Clicking on an individual media title provides you with a brief summary of the story, links to streaming services where you can watch the title, and a number of media titles which are “similar” to it. No explanation is provided about how this similarity is determined.

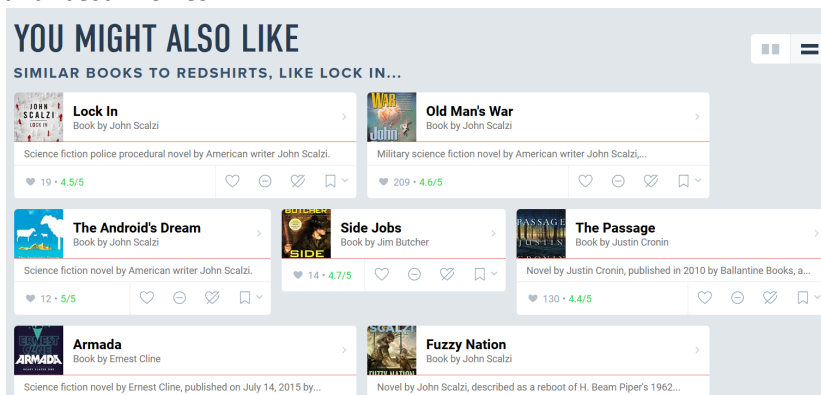
TasteDive succeeds by collecting many different types of media in one place. It allows users to see media titles from a number of streaming services, not just one, and gives users access to them (by

showing where users can stream that title). It also allows users to create lists and view the lists of other users. These are important efficiencies.

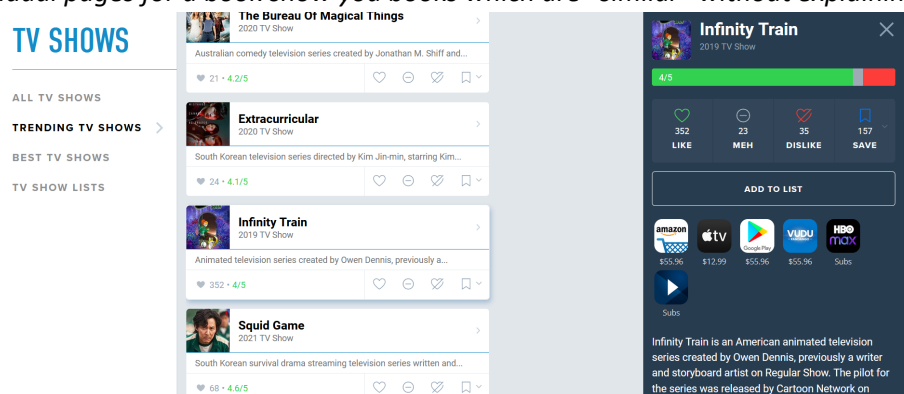
However, like the streaming services, TasteDive does not offer users full control or understanding of their recommendations. The only actual recommendations on the site are based on the undefined characteristics of “trending,” “best,” or “similar.”



TasteDive’s front page takes you to different types of media: ‘everything,’ music, movies, TV shows, books, authors, games, and podcasts. On the ‘Movies’ page, we see user-curated lists containing movies, trending movies, and ‘best’ movies.



Individual pages for a book show you books which are “similar” without explaining why.



TasteDive provides a summary of the media title, the ability to add it to a list, as well as links to where it is available to stream (and for how much money.)

5. Explanation of Design Solution

My design solution is a website called Media Map. There are two central functionalities: ‘recommend’ and ‘media explore.’

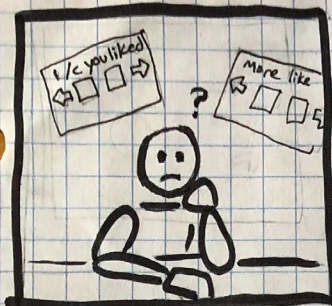
The site's home page is the 'recommend' page. This page shows an image of the user's profile, surrounded by a network-like "map" of media titles, each encircled by a bubble. These are the titles which the AI algorithm determines the user would like most. The titles are linked to the center by lines, each of which is labelled with a particular criterion. The criteria explain why the AI algorithm recommended that media. For example, the criterion "users like you enjoyed this" would mean that this particular title was chosen via collaborative filtering: users who rated other media titles in a similar way to you tended to rate this title highly, which suggests that you will enjoy it too. The criterion "by a creator you like," on the other hand, would mean this title was chosen based on its creator—you tend to like movies by director X, and this movie is also by director X. You can customize the criteria which this page uses.

From the 'recommend' page, you can learn more about the media titles in the bubbles by selecting that bubble.

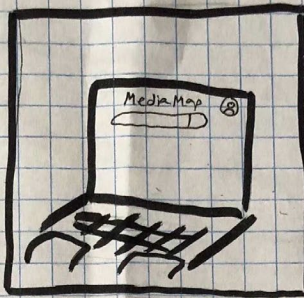
You can also search for a specific media title, or go to it straight from the 'recommend' page. This will take you to the 'media explore' page of that title.

The 'media explore' page has a network-like "map" made of bubbles centered around a specific title. The bubbles contain different attributes of the title: the title's style, its creator, its genre, and so on. Each "attribute bubble" is then linked to other media titles which share these attributes. This functionality allows users to search through content links themselves, entirely on their own terms.

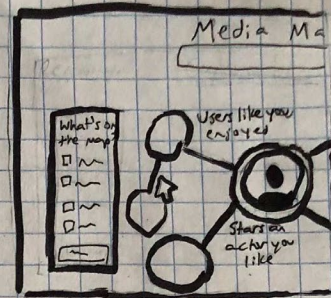
6. *Storyboard(s)*



Ivan doesn't understand why his streaming services recommend certain titles to him. If he doesn't know WHY how can he know if he'll like it??



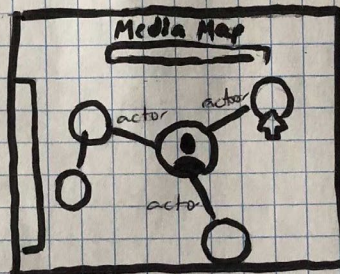
He signs into MEDIA MAP, a site for finding recommendations he can understand.



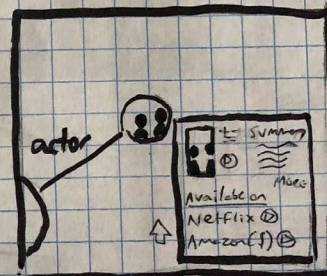
Here, he is directed to different media titles he might like—AND it explains why he might like them!



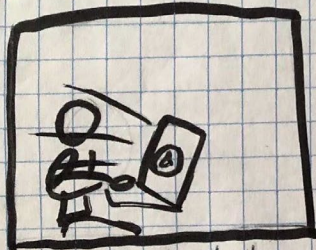
He wants something with the actors he's familiar with, so he changes the settings.



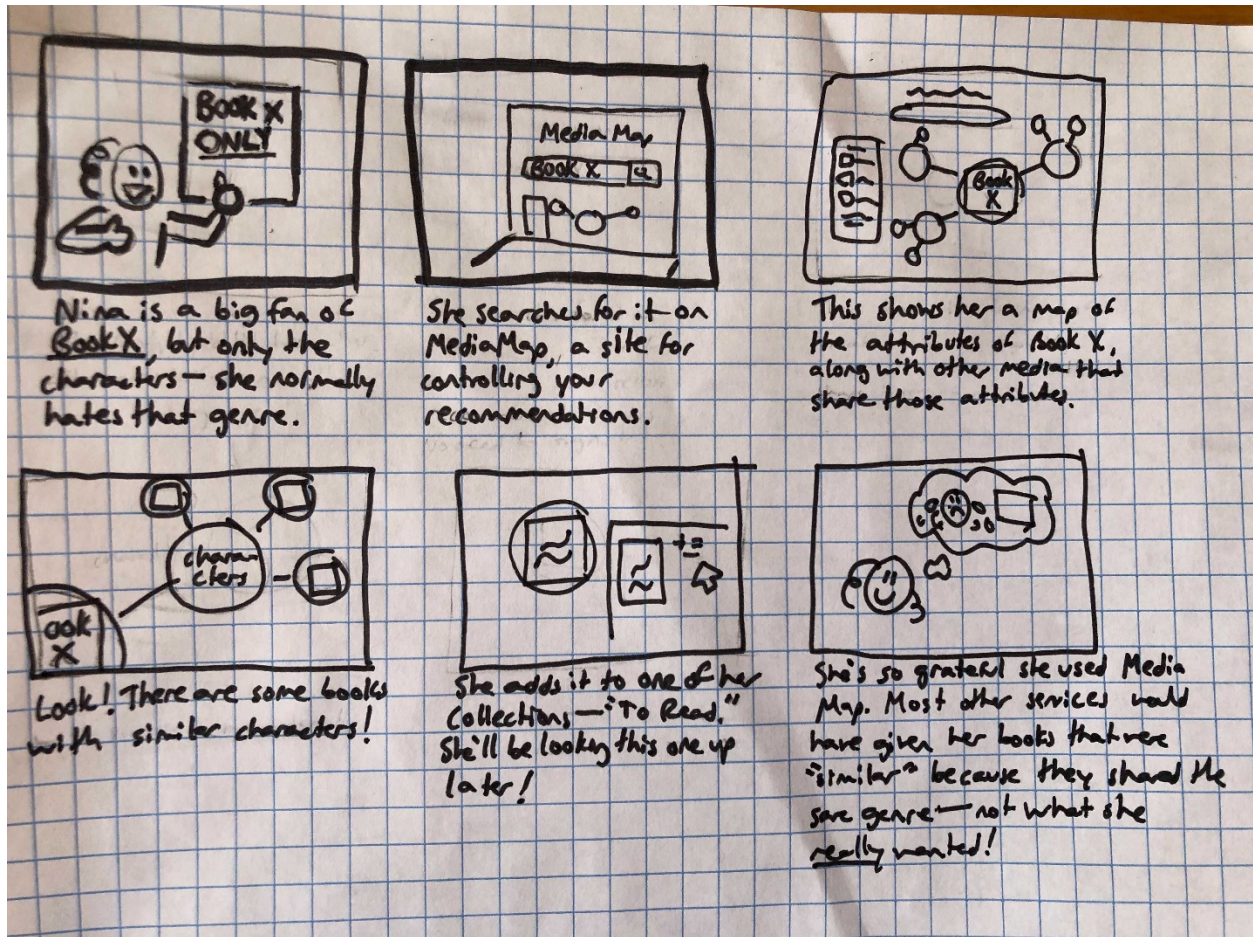
Now he has recommendations the way HE wants! He selects a movie that looks interesting.



He likes the movie's summary and it's available on Netflix!! He clicks to get redirected.



He's ready to start watching!

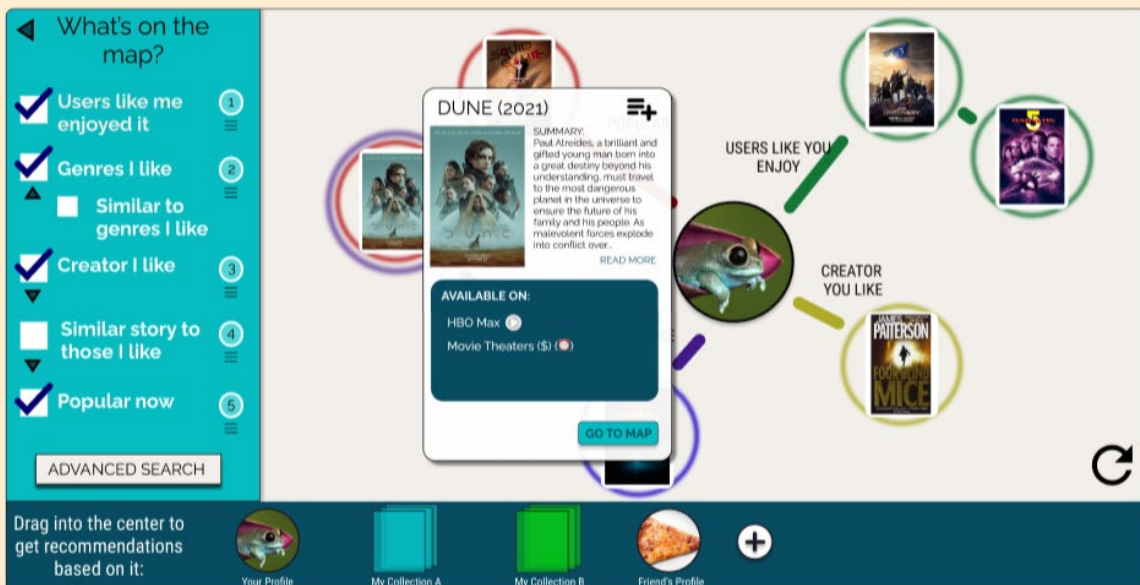


7. Figma Prototype

Media Map



Media Map



Media Map

ADVANCED SEARCH

APPEAL:

POPULAR ✓

CONTENT:

CHARACTER FOCUS ✓

CREATOR:

ACTOR: TIM CURRY ✓

WRITER SIMILAR TO: ISAAC ASIMOV ✓

GENRE:

Search for a specific title

Starcrash



APPEAL:

FAMILY +

CULT CLASSIC ✓

"SO BAD IT'S GOOD" +

INDIE +

GENRES:

SCIENCE FICTION ✓

VINTAGE +

CONTENT:

CREATORS:

USERS LIKE ME LIKED IT:

LOAD SETTINGS

SAVE SETTINGS

CULT CLASSIC ✓

SCIENCE FICTION ✓

CHARACTER FOCUS ✓

POPULAR ✓

ACTOR: TIM CURRY ✓

WRITER SIMILAR TO: ISAAC ASIMOV ✓

Drag items to prioritize.

SEARCH

Media Map



X. Ample User

User Summary:

I am an example user. I like frogs because they hop :)
My favorite phrase is "Lorem ipsum dolor sit amet,"
because it's just so fun to say!
Check out my collections!

Collections:



Watchlist



Viewing History



My Collection A



My Collection B



Friends:



Friend 1



Friend 2



Friend 3



Media Map



Overall Rating



Your Rating



SUMMARY:

A fairy tale adventure about a beautiful young woman and her one true love. He must find her after a long separation and save her. They must battle the evils of the mythical kingdom of Florin to be reunited with each other. Based on the William Goldman novel "The Princess Bride" which earned its own loyal audience.

Predicted Match: 89%

AVAILABLE ON:

- Disney+ (subscription only)
- Amazon Prime Video
- Youtube (\$14.99)
- Apple TV (\$14.99)
- Vudu (\$14.99)

Reviews

Content Warnings

More Info



"Many have said that The Princess Bride is the perfect movie. But after watching it, I'd have to agree. I was surprised by how much I enjoyed it, but the fact is that the movie is charming--and utterly timeless."

- User123



"Honestly? Don't attack me, but it's overrated. There, I said it. The effects are cheesy and the jokes don't land. It's probably better if you have nostalgia, but if you don't? Not worth the watch."

- J. Stapleton



"I love this movie so much! An eternal classic. One of my all-time favorites!"



8. Alternative Implementation

My Individual Design does not include an alternative implementation.

9. Design Process

LOOP 1 (CONCEPTUAL)

Biggest Risk:

The biggest risk in this portion was finding a legitimate need. I had the idea of creating a UI design around finding media content—books, movies, and TV shows. But were people having trouble finding media content? How so? Could I create a design that would fix that problem?

Design:

To mitigate this risk, I did needfinding via contextual inquiry. I had two users take me step-by-step through their media-finding processes. From this, I perceived a need. My users disliked switching between websites to find, learn about, and access visual media, and wanted a better way to find books. There was a need for *convenience*.

In response, I developed two possible design solutions. Then I created two personas of potential users and made storyboards showing how these personas would interact with my designs to solve their problems. Based on these storyboards, I determined the design I would pursue: I would create a hub to concentrate the disparate functionalities of media recommendation, information-gathering, and access.
Evaluation (non-user):

Before prototyping, I spoke to Professor Smith. He suggested that the novelty of my proposed solution was my cross-media recommendations: the fact that I was recommending people movies based on their favorite books, and vice versa. This led me to change my target audience to people who people liked both visual media AND books, not one or the other.

This added a risk to my product: do people who enjoy multiple types of media even want cross-media recommendations? That is, does my product serve the needs of my target audience?

Also, in my studio session, my TA Vishnu said that my idea was not sufficiently original, since it combined existing functionalities without innovating. In response, I added a new functionality to my design: a “content map” visualizing the links between one media title and others.

LOOP 2:

Biggest Risk of Loop:

Once I had a basic design, the biggest risk to tackle was that of user interest. *Will people in my target audience want to use this product?* People will only be interested in my product if it solves the problem of inconvenience that my users had. If it doesn’t solve the problem, then it isn’t useful. Further, is my design sufficiently easy to use that users will find it attractive? If it isn’t usable, then it isn’t convenient—and that means it isn’t useful, either.

Design:

I used the needfinding and storyboards developed in Loop 1 to create an overall design for my product. In order to mitigate the risk of inconvenience, I focused on making the product’s highly usable by keeping ideas such as Nielson’s usability heuristics in mind. I especially concentrated on maintaining consistency with similar platforms and on having an aesthetic and minimalist design.

Prototype:

To handle the risk of user interest, I had to see how my users interacted with all of my product’s functionalities. This way, I could see if the project was sufficiently convenient and interesting for users. To understand user interaction, I made a detailed, low-fidelity paper prototype. The prototype had moving parts which I could use to imitate the experience of using a website. It was also a horizontal, breadth-based prototype. I focused on creating most of the website’s pages so that I could see users interact with all of the website’s features, rather than prototyping one individual feature in great detail.

Evaluation:

I did three user studies examining how users interacted with my paper prototype. I had users do specific tasks and explore the website while giving think-aloud comments, then had them answer interview questions.

The results of the evaluation were surprising to me, and led me to change course entirely. Although the need I had found in my contextual inquiry was real—users did not have an efficient way to combine media-finding functionalities, and found this annoying—users did not find the inconvenience annoying *enough* to change their normal usage patterns. Like my TA, they found most of my project unoriginal.

What interested my users was my last-minute addition of a “content map.” The content map was interesting, new, and solved a *hidden problem*. Users did not perceive an issue with the fact that their media recommendations weren’t based on enough sources (books as well as movies and tv). Instead, their issue was that algorithmic recommendations were unexplained and uncontrollable.

LOOP 3:

New Problem Statement:

With this new user feedback in mind, I changed course. My new problem statement was that users did not have control of their algorithmic media recommendations. They could not understand or control why certain titles were being recommended to them, even though they wanted to do so.

Biggest Risk of Loop:

One of the biggest risks I faced in this loop was fully solving the user problem. In order to allow users to control their recommendations, I had to decide what options to give them. That meant I had to find out what options users wanted. Did they want more options, which would give them flexibility, or very few, for more simplicity? And which options did they want?

The other biggest risk was usability. The purpose of my product is to allow users to understand, visualize, and control their media recommendations. If the recommendations were not formatted in a clear, interesting, and understandable way, it would not fulfill that goal.

Design:

I developed a design to provide users with a “content map.” This “map” would show users what connections existed between different types of media, and allow them to choose what connections they wanted to use to find recommendations.

In order to mitigate risks, I created seven possible designs. This would allow me to check which designs were the best at solving the user problem in a way understandable to users.

Prototype:

I created a non-interactive paper prototype showing my seven possible designs. The paper prototypes were not interactive due to time constraints. When creating the seven possible designs, I mixed my preferred format—a network-like visual map—with other potential designs, including a row-based format similar to that of Netflix and a vertical-column-based format. By mixing these designs, I would see what format users preferred without bias.

I also created a list of possible criteria which users might be interested in when getting media recommendations.

Evaluation:

I showed my paper prototype to three users, and verbally described the user flow.

One user, an avid Netflix user, liked the more Netflix-like format because it was more familiar. The other two users preferred the map-style format *because* it was different from the Netflix format. These users felt that the difference made the design more interesting. By putting my users’ feedback in context, I decided that the map-style format was more appealing to my target audience, which would not have many avid Netflix fans (since these fans would continue to use Netflix.)

Next, I asked users what they thought of the potential links between different media titles. The users indicated that they preferred to have as many potential links as possible. They valued flexibility and control, because they thought they might use a variety of different links.

LOOP 4:

Biggest Risk:

In this loop, my biggest risk was more advanced usability concerns. There are many ways of developing a map-style format. If users could not intuitively understand and navigate the format I chose, the product would not be helpful.

Design:

My design was focused on mitigating the risk of usability. I made my design of the content map more detailed, and created new storyboards to describe the user flow. This was to help me understand how users would interact with the site, and from there make the site easy to use.

Prototype:

I created two more-detailed non-interactive paper prototypes. Again, these prototypes were not interactive due to time constraints. These prototypes demonstrated two separate user flows. One was a more complicated (but more flexible) method for the user to choose what links would be shown on the map. The other was simpler (but more restrictive.)

Evaluation:

I showed my prototypes to two users and explained their respective user flows. The users felt that the overall design was understandable. One user suggested a design with even more flexibility than my own, but the other user preferred the more simple design.

LOOP 5 (Unfinished):

Biggest Risk:

My biggest risk in this loop was big-picture, overall functionality. Was my site a holistic, useful, and interesting website? Was the detailed design unified and understandable?

Design:

Using the user feedback from Loop 4, I improved the usability of my website by turning my more-complicated design into a path for expert users. Most users will see only the more simple design, but I added an “advanced options” button which gives experts a way to further personalize their content map.

In response to the biggest risk of this loop, I created an even more detailed design for my content map to enable me to see the whole of my website.

Prototype:

I created a high-fidelity prototype of my site in Figma. This detailed design will let me see how the website looks as a whole.

Evaluation:

There is no evaluation portion to this loop. This is where the Individual Project ends.

10. *Future Work*

The biggest risk that I currently perceive is that of user breadth. In the course of the project, I was only able to speak with six users. Most of these users were Barnard or Columbia students—meaning that they all come from a very similar context—and all of them shared traits relevant to my imagined target audience—they were interested in both books and visual media. I do not think this number of sample users is sufficient to determine whether my project meets a widespread user need. I also don’t think it is sufficient for determining whether the format of the project is successful at meeting that need for a wide variety of people. My central unknown, then, is whether the needs and desires of *these* users are the same as those of *many* users.

Also, I want to expand user breadth in order to re-evaluate my target audience. Obviously there *are* people who enjoy both books and visual media—I found six of them. However, I do not know if that is the best audience to target. Far more people want to find visual media only, and they may also want a way to understand and control their recommendation. Would my project be more useful if it was aimed at this larger target audience?

To mitigate this risk, I would interview more users, and analyze their response data in a more systematic way. In this way, I would know whether my target audience is relevant, and whether my project is appealing—in premise and in design—to a wider variety of people from different contexts.

11. *Post-mortem*

What went best in this project was probably the stroke of serendipity of the ‘content map’ idea. Although it was a last-minute addition to the prototype, users’ positive reactions to it gave me a new direction when my original direction failed. That, of course, leads into what went wrong—namely, my original project idea. Most obviously, the idea was unoriginal. I wasn’t adding any functionality, merely putting existing functionalities in the same place, and that wasn’t enough to interest users. Further, in retrospect, I see that I created my solution without thoroughly understanding the problem *as it was perceived by users*. In my contextual inquiry, I saw that the media-finding process was inconvenient and annoying to users. However, I didn’t sufficiently check the perception of the users about it. Users were annoyed, but not annoyed enough to change their process. If I were to undertake the project again, I would change how I did my contextual inquiry. I would work harder to minimize my assumptions about how users should react. I started off trying to fix a mostly-acceptable inefficiency because I was looking for where *I* saw a problem, and instead, I would try to find where *my users* saw a problem.

Alternatively, if I were to do the project again, I might change the nature of the idea as a whole. My user population included myself, and it would be interesting to do a design further afield from my own knowledge.

12. *Optional permission to feature your project in Prof. Smith’s teaching*

I, Adi, give Prof. Brian A. Smith permission to feature this project in his teaching materials. Prof. Smith may use my name when describing my project.